HFT for Dummies, Part 1: Basic Intro to Algo-Trading

Tristan (@Tristan0x), October 28, 2026
https://x.com/Tristan0x/status/2115509914768687299

How I made $6K PnL in 30 days on $10K principal market making perps ($158m+ volume).

The strat was entirely built by me and my quant, Claude (he won a national math competition in China).

Last night I asked if you'd want a guide. Hundreds of you said yes, so here it is: HFT for Dummies.

The honest timeline when I did it:
- Days for basic harness and venue integrations.
- Weeks to get it semi-competitive.
- Months to really tune and get it properly profitable.

This series can hopefully shortcut the learning curve for you.

I've been writing trading bots since 2017. It's how I got into crypto (free market data!), and how I met my co-founder.

Back then the barrier was high. You had to know how to code, to study data to find alpha, and to grind devops to have systems not blow up in production. Thankfully I taught myself Python and later Rust, but not everyone has a CS degree.

Nowadays with LLMs, anyone can have a skilled quant in their pocket. I use it for data analysis, strategy tuning, research and alpha generation and infra optimisation. It's a ridiculous time saver.

I'm extremely bullish on one-person quant shops. I think this will be a huge trend this next decade as finance comes on chain and market data and connectivity are accessible to anyone. I've already met many hobbyists who've made a living trading on e.g. Hyperliquid, enough so to quit their day jobs.

What we'll be building

A delta-neutral, reference-price market maker.

Binance is where most price discovery happens. We stream its market data, use it as a reference price, and quote a bid and an ask around it on another venue. I'm using @Bulletxyz because it's the one I know best.

Market making is the business of picking up pennies. Mine are under half a basis point per dollar. Enough pennies add up at the scale of millions of volume.

The catch is speed. If the market moves and you're slow to update, someone lifts your stale quote and you're the exit liquidity. It's a fun game of algorithmic cat and mouse.

On Bullet, takers get a 100ms speed bump. Why? It gives soon-to-be market makers like you a safety buffer so you don't get steamrolled by billion dollar firms with microwave towers.

The goal

The goal: tick to order sent in under 10ms. Stretch goal: under 1ms, which is very achievable if you're determined. We'll also focus on minimising jitter (randomness) because your slowest orders are the ones that will lose you the most pnl.

Agenda:
I'll aim to post one topic a day, going from easy wins, and getting progressively more deep in the weeds. If you work at any HFT firm this will all be elementary stuff, but hopefully it's helpful for newcomers.

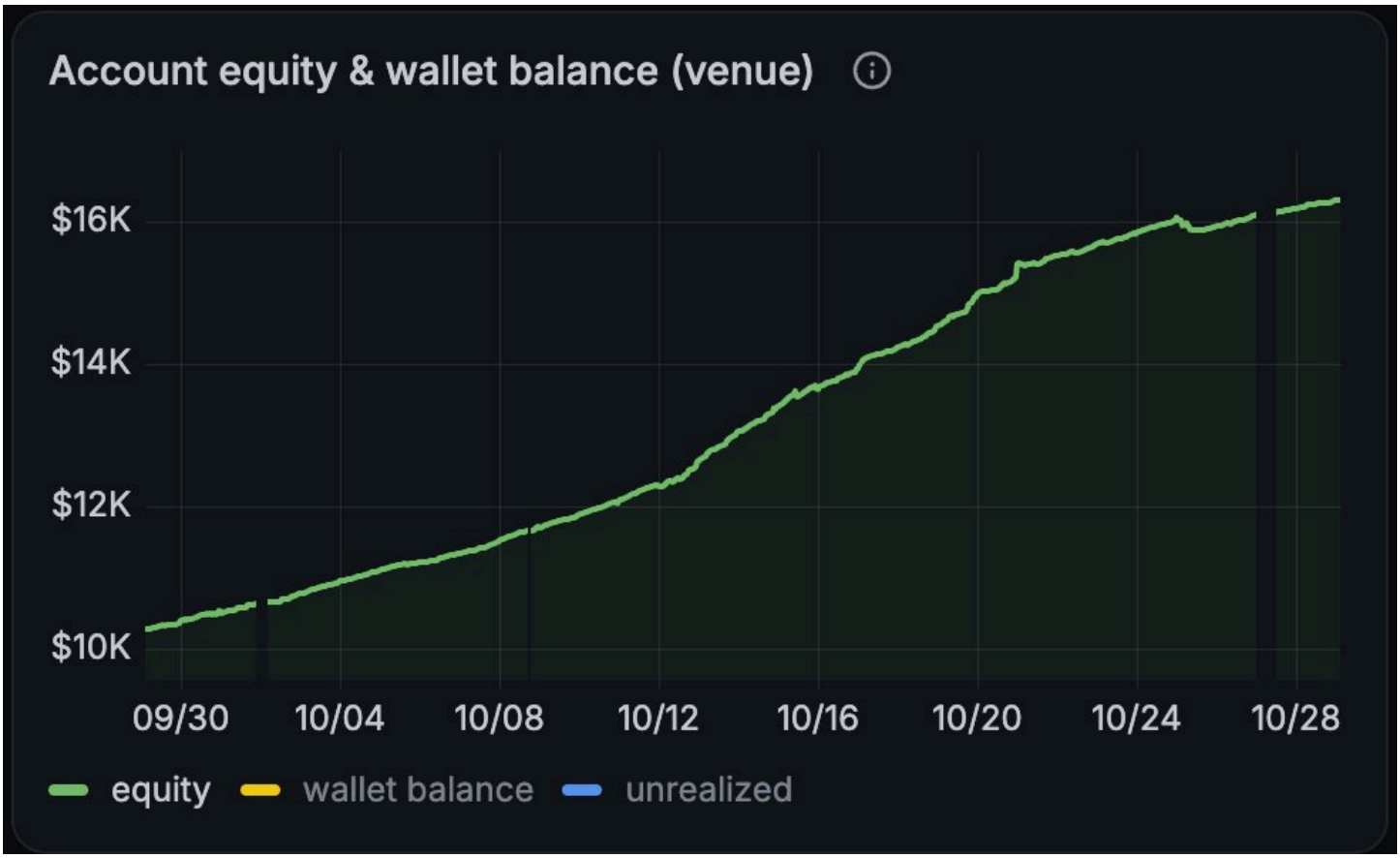
Saving seconds
1. Basic intro to algo-trading (this post)
2. Running a simple market maker from your laptop
3. Measure once, cut twice: telemetry and Amdahl's law
4. AWS: Infra and colocation

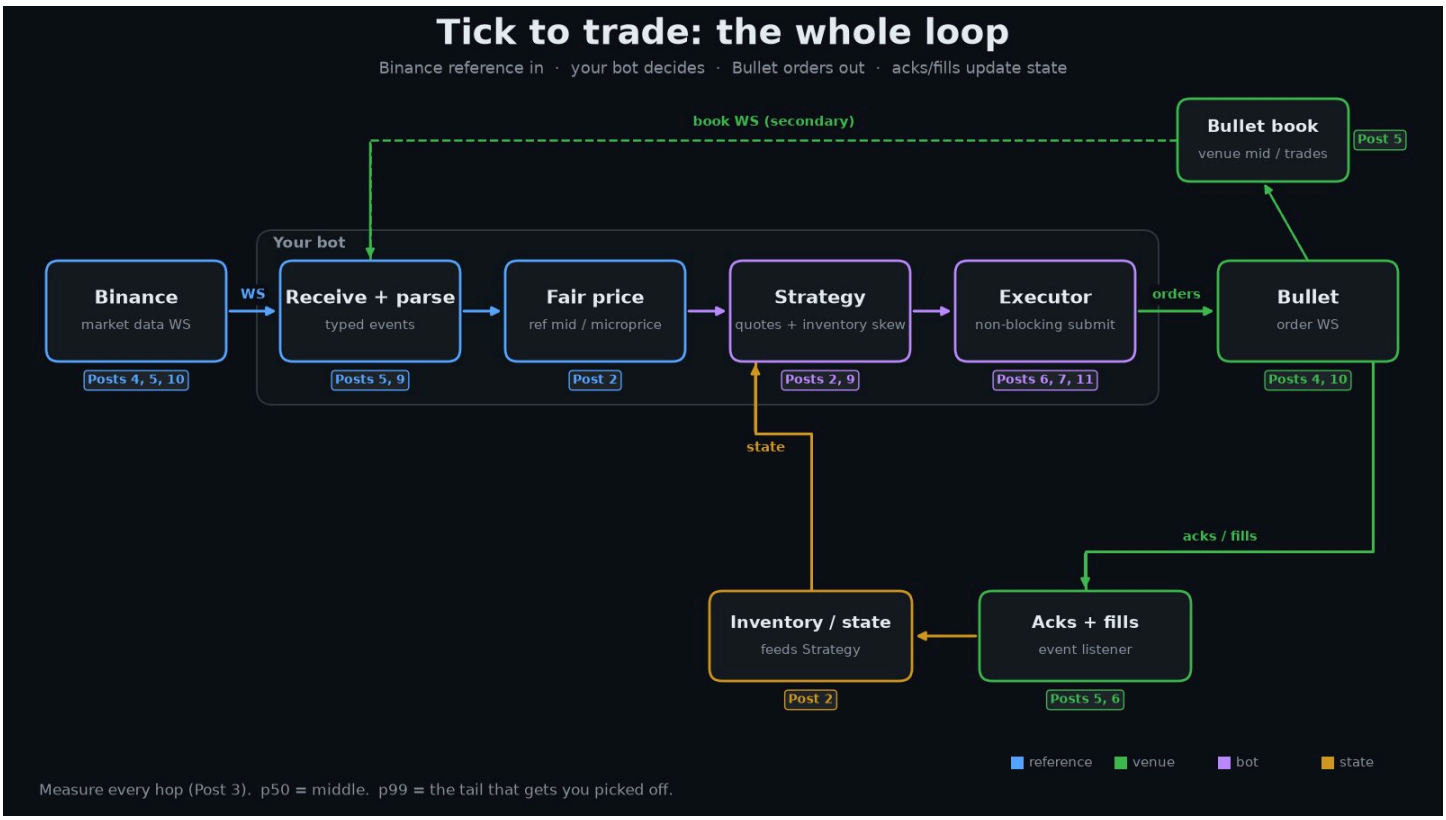
Saving millis
5. Stop polling: websockets are your friend
6. Optimising order execution
7. How to not get rekt by API rate limits
8. Linux tuning: CPU pinning, IRQs and other nerd stuff

Saving micros (diminishing returns, but fun)
9. Optimising the hot path with CPU profiling in Rust
10. AWS: the last mile
11. Bonus: nonces (and other DeFi quirks)

I'll try share some open source code along with it. Don't expect to be beating Jump in the orderbooks just yet. It should be a good baseline that you can extend and take it as far as you want (you can take it pretty far!).

Tomorrow: setting up a basic MM in Rust, the architecture of which will roughly follow the diagram attached below.


Account equity & wallet balance (venue)
$16K
$14K
$12K
$10K
09/30
10/04
10/08
10/12
10/16
10/20
10/24
10/28
equity
wallet balance
unrealized


Tick to trade: the whole loop
Binance reference in · your bot decides · Bullet orders out · acks/fills update state
book WS (secondary)
Bullet book
venue mid / trades
Post 5
Your bot
Binance
market data WS
Posts 4, 5, 10
WS
Receive + parse
typed events
Posts 5, 9
Fair price
ref mid / microprice
Post 2
Strategy
quotes + inventory skew
Posts 2, 9
Executor
non-blocking submit
Posts 6, 7, 11
orders
Bullet
order WS
Posts 4, 10
state
acks / fills
Inventory / state
feeds Strategy
Post 2
Acks + fills
event listener
Posts 5, 6
reference
venue
bot
state
Measure every hop (Post 3). p50 = middle. p99 = the tail that gets you picked off.